

Expressions, Formulae and Brackets — Paper A

MEDIUM · NON-CALCULATOR · 40 MINUTES · 30 MARKS

NAME

DATE

SCORE / 30

- Answer every question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- No calculator. Every number on this paper is small enough to handle on paper.
- When you substitute a value, write it in brackets: $3(-2)$, not $3-2$. Most lost marks on this paper are dropped signs.
- Only like terms can be added — terms with exactly the same letter part. $3x + 2$ is already simplified.
- “Factorise completely” means take out the highest common factor. Check by looking inside your bracket: if the numbers there still share a factor, you stopped early.
- The mark for each question is shown in square brackets on the right.

SECTION 1 – SIMPLIFYING AND WRITING EXPRESSIONS

1. Complete the table. Each empty box is worth one mark. [6]

Expression	Simplified	Its value when $x = 2$
$5x + 3 - 2x + 7$		
$8x - 3 - 5x + 9$		
$4x + 2x - 7 + 1$		

Underline each term together with the sign in front of it before you start. The sign always travels with its term.

2. Simplify:

(a) $9p + 4q - 5p + 2q$ [1]

(b) $6m \times 4n$ [1]

(c) $3c \times 5c$ [1]

3. Using n for the number, write an expression for each of these.

(a) five more than three times the number [1]

(b) the number decreased by seven [1]

(c) double the sum of the number and four [1]

SECTION 2 – SUBSTITUTING, EXPANDING, FACTORISING

4. Given that $a = 5$ and $b = -3$, work out:

(a) $2a + b$ [1]

(b) $3b - a$ [1]

(c) ab [1]

(d) $a - b$ [1]

5. Expand:

(a) $3(x + 4)$ [1]

(b) $5(2x - 3)$ [1]

(c) $4(3x + 2)$ [1]

(d) $2(5 - x)$ [1]

6. Factorise completely:

(a) $6x + 9$ [1]

(b) $10x - 15$ [1]

(c) $8x + 20$ [1]

(d) $14x + 21$ [1]

SECTION 3 – USING IT

7. A taxi charges a fixed 40 rupees, plus 12 rupees for every kilometre travelled.

(a) Write an expression for the cost of a journey of d kilometres. [1]

(b) Find the cost of a journey of 8 km. [1]

(c) Find the cost of a journey of 15 km. [1]

8. A rectangle has width w cm and length $(w + 6)$ cm.

(a) Write an expression for the perimeter, fully simplified. [1]

(b) Find the perimeter when $w = 5$. [1]

(c) Write an expression for the area, with the bracket expanded. [1]

End of paper. Check every answer has a unit or a form the question actually asked for.